

Lab 4 — Minesweeper

LLM Prompts — Tasks 1 to 4 only

Prompt 1 — Task 1: Boundary-Safe Board Traversal

I am working on a Minesweeper assignment.

Please inspect the existing main.py, game.py and board.py that I provide.
Do not rewrite the project or change unrelated functionality.

Task 1:

Correct the neighbour traversal so that every coordinate visited by reveal and mine-counting logic is a valid board coordinate.

Requirements:

- Flood-fill must work correctly from corners, edges and interior cells.
- No invalid, negative, out-of-range or phantom coordinates should be visited.
- Keep the existing modular structure.
- Do not add external dependencies or persistent storage.

First explain the existing defect and why it occurs.
Then make the minimal code changes required to fix Task 1.

Prompt 2 — Task 2: Complete Win and Flag Behaviour

Now implement Task 2 only.

Requirements:

1. Keep the existing flag command.
2. Flags must be toggleable.
3. Revealing a flagged cell must not accidentally change its state.
4. Invalid actions must not corrupt the board state.
5. The game must recognise a win only when every non-mine cell has been revealed.
6. Keep the existing project structure and commands.
7. Do not add persistence or unnecessary dependencies.

First explain the current flag and win logic.
Then make the minimal changes required to satisfy Task 2.

Prompt 3 — Task 3: Difficulty Modes

Now implement Task 3 only.

Add Easy, Medium and Hard difficulty modes.

Requirements:

- Each mode must have different board dimensions and mine counts.
- Difficulty selection must happen entirely in memory.
- Do not create CSV, JSON, SQLite or any other data files.
- All existing reveal, flag and command behaviour must continue to work.
- Keep the existing modular structure.
- Do not rewrite unrelated code.
- Do not add unnecessary external dependencies.

First explain where difficulty should be represented in the current code.
Then make the minimal changes required.

Prompt 4 — Task 4: Action-Level Feedback

Now implement Task 4 only.

Add concise player-facing feedback for a reveal action.

Requirements:

- One user reveal command counts as one player action.
- A flood-fill may internally process many cells.
- Internal flood-fill iterations must NOT produce repeated player-facing feedback.
- Feedback should happen once after the reveal command completes.
- Keep the existing commands and modular structure.
- Do not rewrite unrelated functionality.

First explain where the feedback should be generated and why.

Then make the minimal code changes required.

Note: These are the four task-specific prompts only, corresponding to Tasks 1–4 in the Minesweeper assignment.